

Leonardo Marciaga

3201 South State Street, Chicago, IL, Mailbox 2321
(872) 279-9766 • leonardomarciaga@gmail.com • leonardo-m12.github.io

EDUCATION

BS, Applied Mathematics

Expected May 2027

Illinois Institute of Technology

Minors in Artificial Intelligence and Statistics

- Cumulative GPA: 4.0/4.0
- Relevant coursework: Modern Methods in Discrete Applied Mathematics (graduate), Data Mining, Graph Theory, Data Structures and Algorithms, Linear Optimization (ongoing), Introduction to Stochastic Processes (ongoing)

RESEARCH EXPERIENCE

Research Fellow

January 2026 - Present

SoReMo Initiative, Illinois Tech

- Ongoing original research under the Socially Responsible Modeling, Computation and Design (SoReMo) Initiative at Illinois Tech.
- Developing a web-based tool using discrete-event simulation to improve client flow at the local St. James Food Pantry. The project's goal is to enable pantry managers to explore the effect of different client flow process structures and recognize improvement areas via an accessible interface to a simulation model.

Research Intern

June - August 2025

Institute for Pure and Applied Mathematics (IPAM), UCLA

- Collaborated with the RAND Corporation in development of methodology leveraging Deep & Cross neural network architectures, LLM embeddings, and cluster-informed data generation to address challenges of survey evolution and data sparsity in the social sciences.
- Developed and validated LLM-based synthetic data generation pipeline against data quality metrics, achieving 80% overall similarity to real-world statistical trends.
- Applied clustering algorithms such as K-medoids, hierarchical, and spectral clustering, in complex longitudinal survey data encompassing nearly 10 years and 2000+ respondents to characterize generally positive/negative attitudes towards vaccination. This was used to supplement synthetic data generation.
- Accepted for publication at the 2026 AAAI Student Abstract and Poster Program as listed below.

Research Assistant

August 2024 - Present

Department of Applied Mathematics, Illinois Tech

- Ongoing project with Professors Hemanshu Kaul (Illinois Tech) and Jeffrey Mudrock (University of South Alabama) on graph coloring applying recent methods from probability, algebra, and matching theory.
- Resulted in the solution of an open question in graph coloring using methods from graph criticality, enumerative functions and real analysis including Karamata's inequality. Resulted in a published at an international peer-reviewed journal as listed below.

PUBLICATIONS

Rezvani, J., Hyk, A., Pham, T., Marciaga, L., Liao, C., Vardavas, R., Mitsopoulos, K. *Semantic Embedding and Synthetic Augmentation for Longitudinal Survey Prediction (Student Abstract)*. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence (AAAI-26)*. Singapore. (in press).

Kaul, H., Marciaga, L., and Mudrock, J. *List Coloring the Cartesian Product of a Complete Graph and Complete Bipartite Graph*. Enumerative Combinatorics and Applications, 6 (2026), article S2R5. Available at doi.org/10.54550/ECA2026V6S1R5.

TEACHING

Associate Instructor

January 2021 - Present

Panamanian Mathematical Olympiad Foundation, Panama

- Led training sessions for Panama’s math contest training program, leading to award-winning students at contests such as the International Mathematical Olympiad, the Ibero-American Mathematical Olympiad, and the Pan-American Girls’ Mathematical Olympiad.
- Mentored students, organized training schedules and created class material in mathematical olympiad topics, such as algebra, combinatorics, geometry, and number theory.

Teaching Assistant

August - December 2024

Department of Computer Science, Illinois Tech

CS 105: Introduction to Computer Programming

- Taught basic programming concepts to 15 non-CS students, providing personalized assistance and feedback, both in-class and through homework.
- Identified areas for improvement in class performance through discussions with lead instructor and other teaching assistants.

LEADERSHIP

President of SIAM IIT Student Chapter

January 2026 - Present

Illinois Tech

- Recently elected president of the SIAM IIT Chapter. Working on several initiatives for increasing member engagement, such as a trip to Argonne National Lab and events with Wolfram Research.

Secretary of SIAM IIT Student Chapter

January 2024 - December 2025

Illinois Tech

- Write the chapter’s weekly newsletter for 100+ students and faculty, with average click rate of 36%.
- Lead new training seminar series for the Putnam Competition, presenting problem-solving techniques and discussing solutions to past competition problems.
- Volunteered as desk assistant for the SIAM Great Lakes Section Annual Meeting with 150+ attendants.
- Designed new chapter website with HTML and CSS.

HONORS AND AWARDS

AAAI-26 Student Scholarship & Volunteer Award	2025
Illinois Tech Clinton E. Stryker Service and Leadership Award	2025
85th William Lowell Putnam Mathematical Competition - Top 500	2024
International Mathematical Olympiad (IMO) - Honorable mention	2018 - 2020

PROFESSIONAL MEMBERSHIPS

Association for the Advancement of Artificial Intelligence (AAAI)	2025 - Present
Society for Hispanic Professional Engineers (SHPE)	2025 - Present
Society for Industrial and Applied Mathematics (SIAM)	2024 - Present

SKILLS

Programming: Python, R, HTML, CSS, MATLAB, SQL, Mathematica
AI/Machine Learning Libraries: scikit-learn, TensorFlow, PyTorch
Data Analysis & Visualization Libraries: pandas, matplotlib, tidyverse, ggplot2, plotly